
PH 705: Foundations in Experimental Physics

Course Code	PH 705
Course Name	Foundations in Experimental Physics
L-T-P-C	3-0-1-4
Intended for	Ph. D (Physics)
Distribution	Elective
Approval	1st Adhoc Senate (Apr 2011)

Course Contents

- **PART-A** Lectures on experiments, which made an impact on physics and/or lectures on new/modern experiments of current importance to frontier research in Physics. (Lecture Hours: 24)
- A1: Great Experiments In Physics:
- Starting with Galileo's experiments with motion, this study of 25 crucial discoveries includes Newton's laws of motion, Chadwick's study of the neutron, Hertz on electromagnetic waves, and more.
- A2: Top 10 beautiful experiments:
 1. Young's double-slit experiment applied to the interference of single electrons
 2. Galileo's experiment on falling bodies (1600s)
 3. Millikan's oil-drop experiment (1910s)
 4. Newton's decomposition of sunlight with a prism (1665-1666)
 5. Young's light-interference experiment (1801)
 6. Cavendish's torsion-bar experiment (1798)
 7. Eratosthenes' measurement of the Earth's circumference (3rd century BC)
 8. Galileo's experiments with rolling balls down inclined planes (1600s)
 9. Rutherford's discovery of the nucleus (1911)
 10. Foucault's pendulum (1851)
- Others experiments:
 - Archimedes' experiment on hydrostatics
 - Roemer's observations of the speed of light
 - Joule's paddle-wheel heat experiments
 - Reynolds's pipe flow experiment
 - Mach & Salcher's acoustic shock wave
 - Michelson-Morley measurement of the null effect of the ether
 - Röntgen's detection of Maxwell's displacement current
 - Oersted's discovery of electromagnetism
 - The Braggs' X-ray diffraction of salt crystals
 - Eddington's measurement of the bending of starlight
 - Stern-Gerlach demonstration of space quantization
 - Schrödinger's cat thought experiment
 - Trinity test of nuclear chain reaction
 - Wu et al.'s measurement of parity violation

- Goldhaber’s study of neutrino helicity
- Feynman dipping an O-ring in water
- A3: Experiments of current interest:
 - A3.1 Proton lifetime measurement
 - A3.2 Bose-Einstein Condensation.
 - A3.3 Measurement of the Fine-Structure Constant.
 - A3.4 Experimental tests of Bell’s inequalities
 - A3.5 Experiments on Quantum Computation
 - A3.6 The High Temperature Superconductivity Space Experiment
 - A3.7 Study of electronic structure of materials using photoemission spectroscopic experiments (angle integrated, angle resolved and spin resolved photoemission experiments)
- Part-B Actual Laboratory Experiments [14 Lecture Hours]. Experiments will be chosen from the list below:
 1. Four probe method
 2. Michelson Interferometer (white light)
 3. Sand piles and rice piles, avalanche distribution
 4. X-ray of an NaCl single crystal
 5. Directed percolation – spreading of ink on paper
 6. Viscous fingering - effect of viscosity
 7. Hall effect
 8. Measurement of Band Gap in a Semiconductor
 9. Construction of a hologram
 10. Zeeman Effect
 11. Kerr Effect
 12. Shot Noise and Johnson noise – measurement of Boltzmann Constant
 13. Absolute zero of temperature and charge of electron
 14. Preparation (CVD) and characterization (AFM, STM) of thin films
 15. Experiments on photon squeezing, Bose-Einstein condensation, parity-violation in weak interaction
- Part – C Demonstrations in the experimental laboratories in our Institute [Lecture Hours: 18]

References

1. Morris H. Shamos, **Firsthand Accounts from Galileo to Einstein**
2. <http://physicsweb.org/articles/world/15/9/2>
3. Webb, Denis C., Nisenoff, M., **Microwave Journal**, 1991.
4. Nisenoff M., Gubser D.U., Wolf, S.A., Ritter J.C., Price G. Source, **High temperature superconductivity space experiment (HTSSE)**, Superconductor, Science and Technology, Volume 4, Number 9, 1991.
5. Stephan Hufner(ed.), **Very high resolution Photoelectron spectroscopy**, 2007.